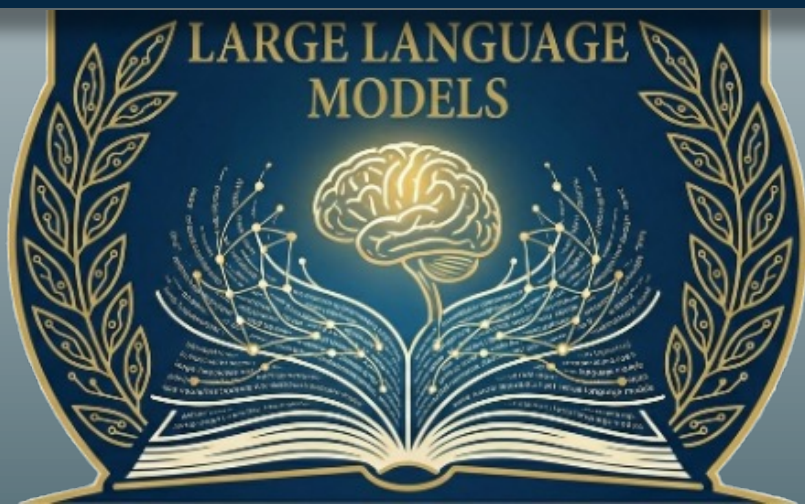


Instruction Tuning



Mohammad Javad Dousti

Overview

❑ Pre-training (training)

- Self-supervised learning (SSL)
- Big data ↔ Big model
- Objective: Next token prediction
 - Learns grammar
 - Learns language nuances
 - Learns general knowledge and trivia

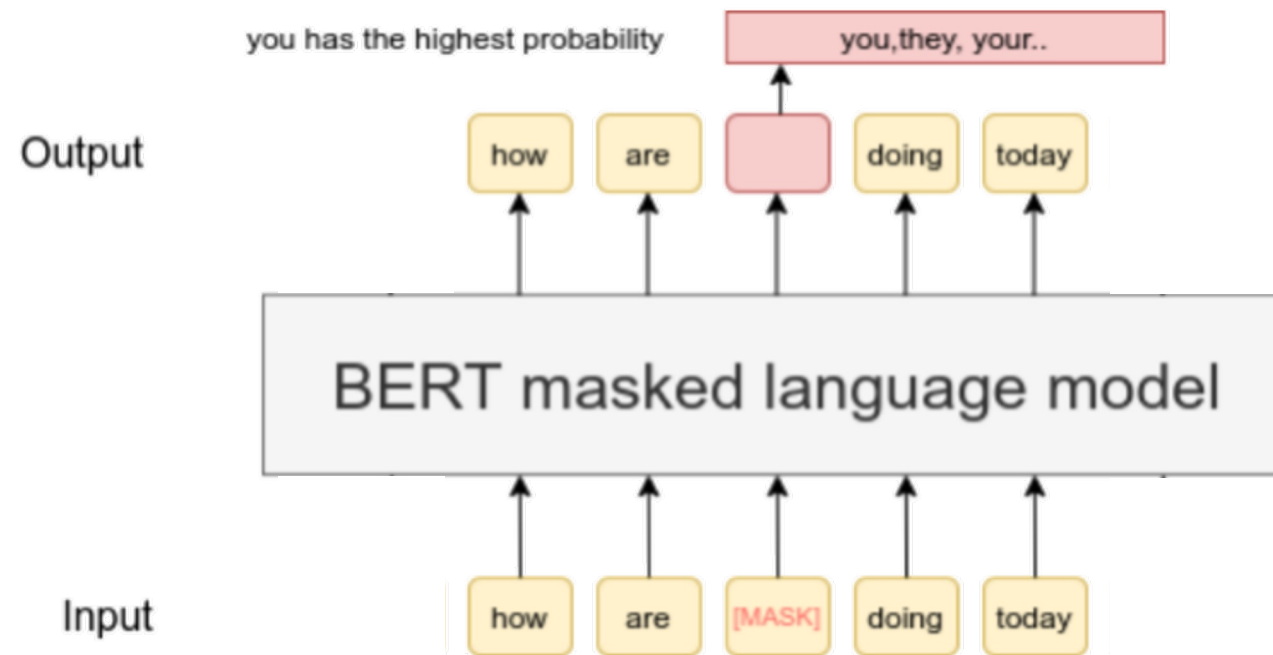
❑ Fine-tuning

- Supervised
- Small data
- One or several domains



Encoder Only (1)

- ❑ How to prevent cheating? Next word prediction fundamentally doesn't work for bidirectional models, instead do masked language modeling.
 - Hides some words, and then tries to predict the hidden words
- ❑ BERT formula: Take a chunk of text, mask out 15%-30% of the tokens, and try to predict them.



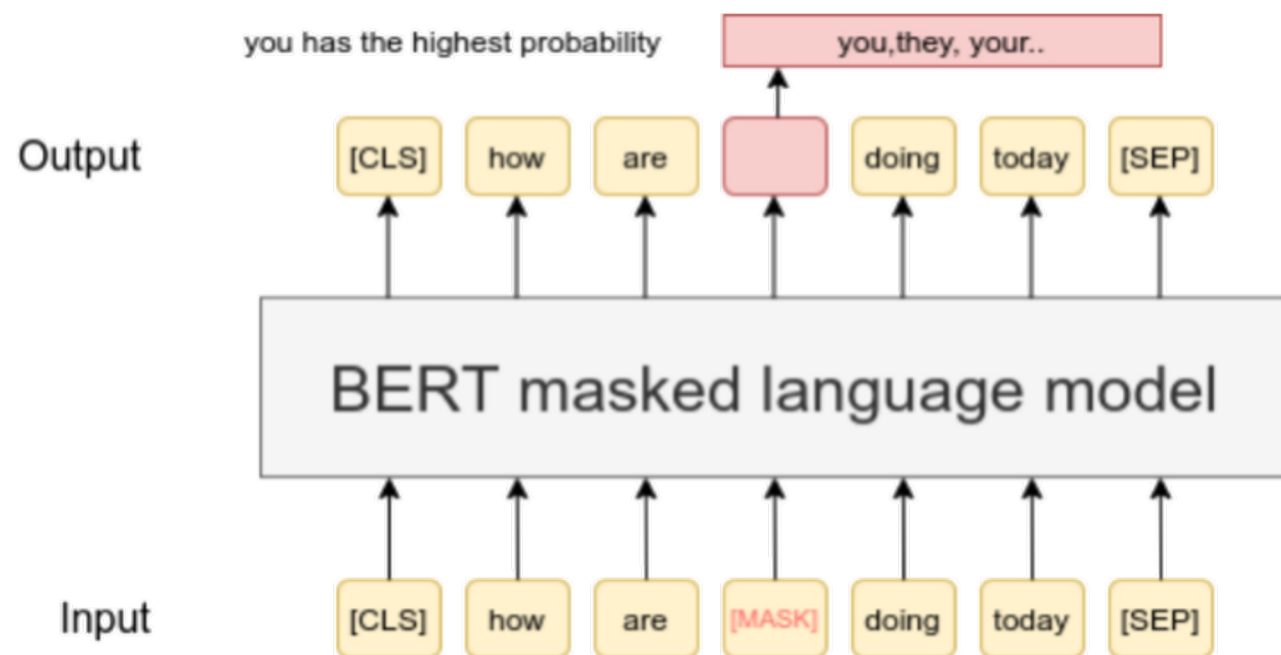
Encoder Only (2)

□ Next sentence prediction task

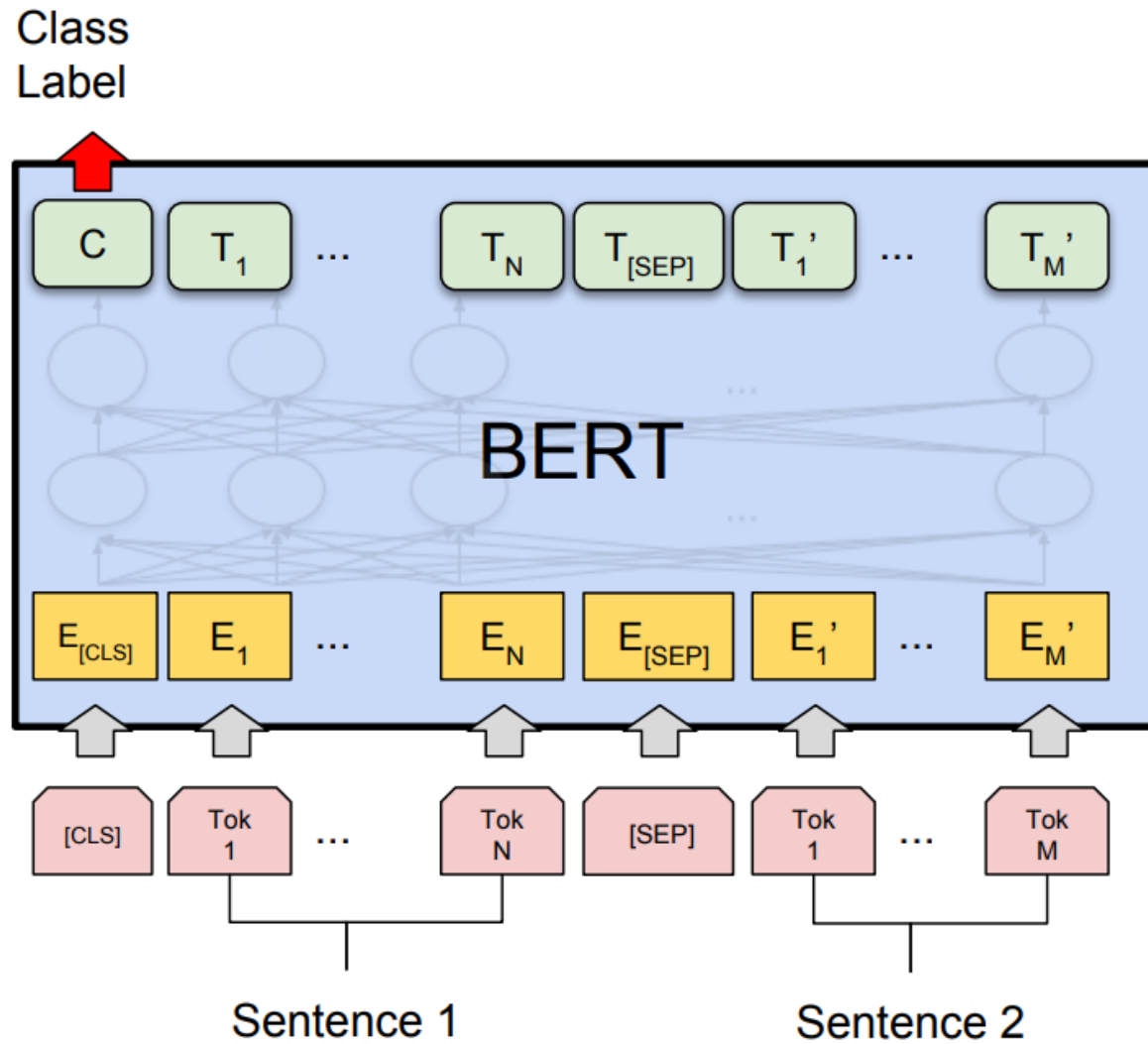
- Given 2 sentences, the model learns to predict if the 2nd sentence is the real sentence, which follows the 1st sentence. This is done through the [CLS] and [SEP] token.

□ [CLS] appears at the very beginning of each sentence, it has a fixed embedding and a fix positional embedding, thus this token contains no information itself.

- However, the output of [CLS] is inferred by all other words in this sentence, so [CLS] contains all information in other words.
- [CLS] is a good representation for sentence-level classification.
- $\text{Softmax}(W_{CLS} \cdot h_{CLS}^T)$ dimension?

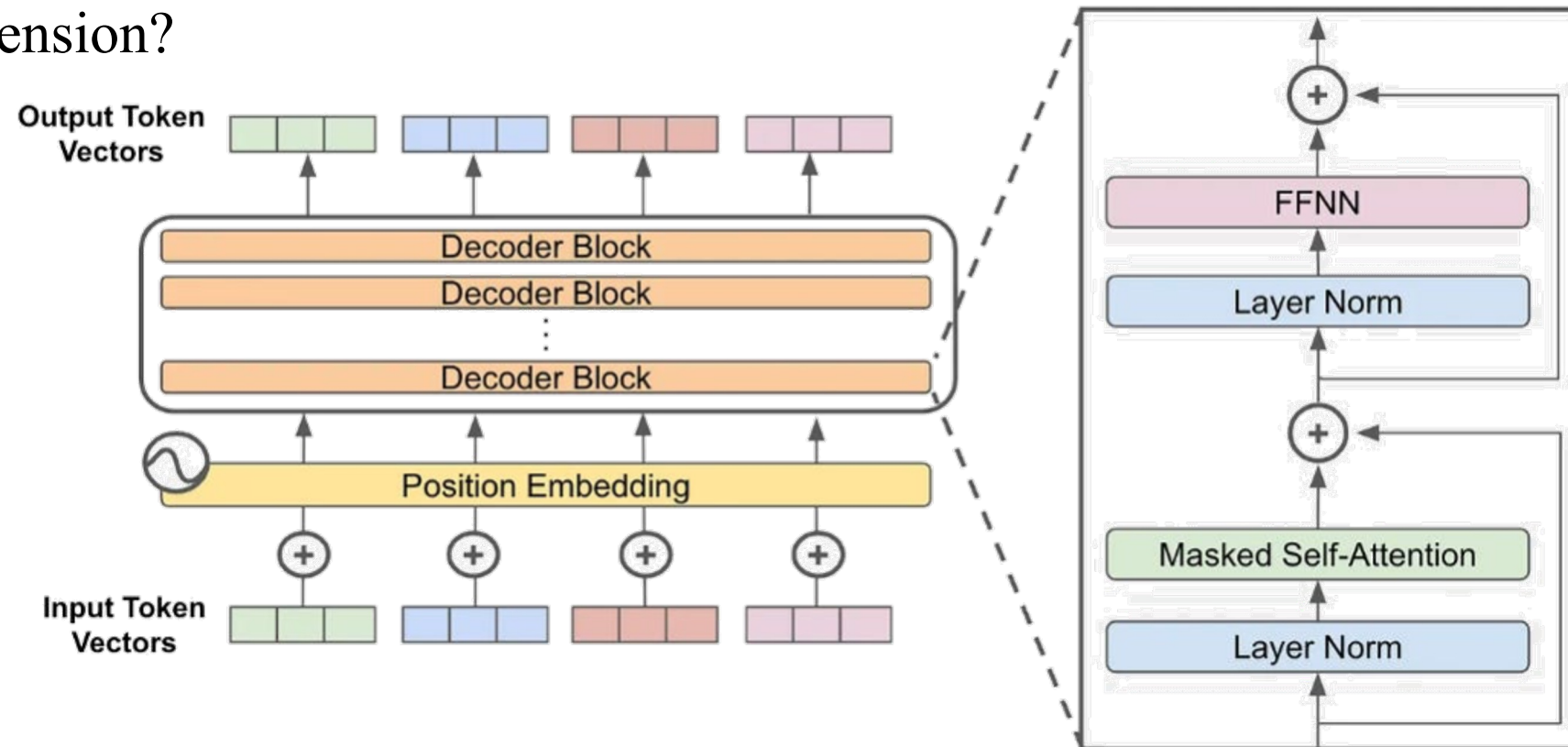


Encoder Only (3)



Decoder only

- ❑ Partial masking
- ❑ Autoregressive.
 - Teacher forcing (during training only)
 - $\text{Softmax}(W \cdot h^T)$ dimension?

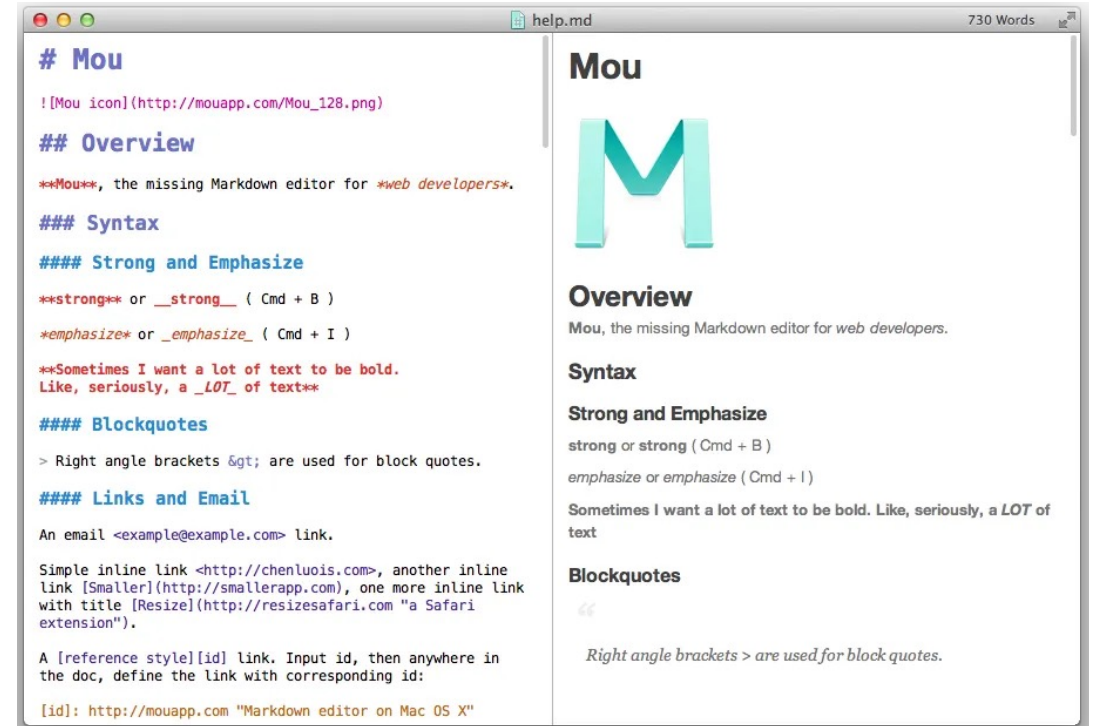


Instruction Tuning

❑ Also called fine-tuning or supervised fine-tuning (**SFT**)

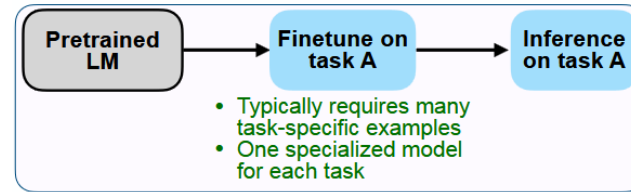
❑ Instruction

- Question + Response
- **Count:** a few thousands (much smaller than pre-training data)
- **Cost:** **high** (human curated)
- **Format:** Markdown or LaTeX (especially for equations, e.g., $\$2^5\$$)
- Variety matters, a **LOT**

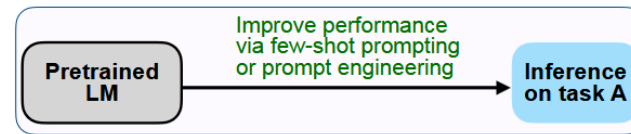


Finetuned Language Models Are Zero-Shot Learners (1)

(A) Pretrain–finetune (BERT, T5)



(B) Prompting (GPT-3)



(C) Instruction tuning (FLAN)

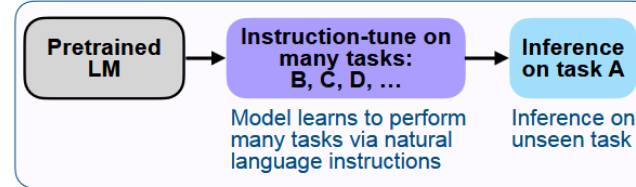


Figure 2: Comparing instruction tuning with pretrain–finetune and prompting.

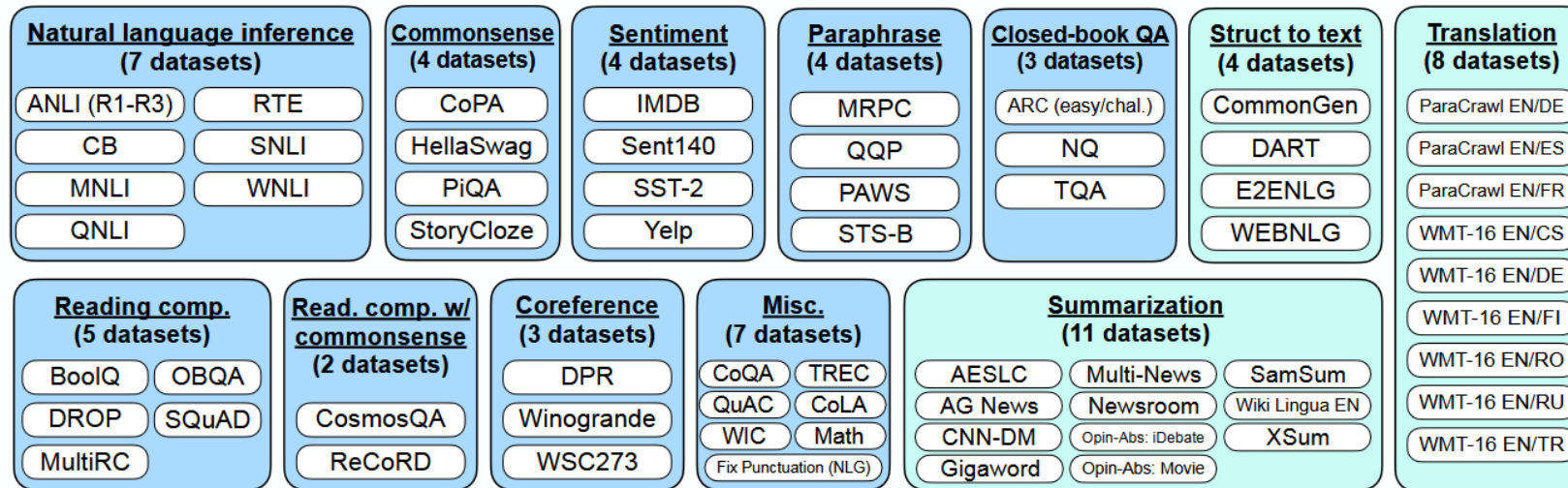


Figure 3: Datasets and task clusters used in this paper (NLU tasks in blue; NLG tasks in teal).

Finetuned Language Models Are Zero-Shot Learners (2)

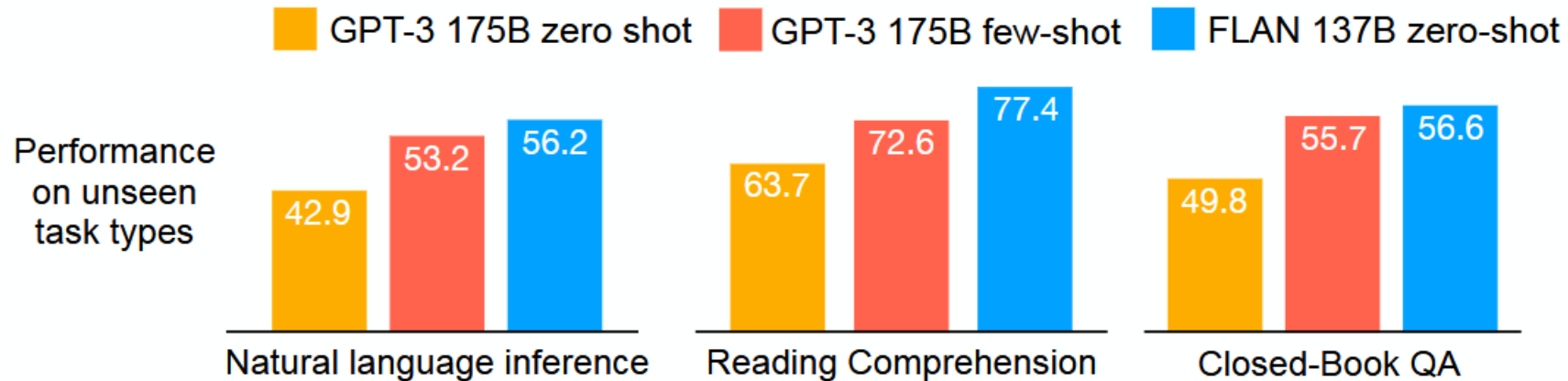


Figure 1: Top: overview of instruction tuning and FLAN. Instruction tuning finetunes a pretrained language model on a mixture of tasks phrased as instructions. At inference time, we evaluate on an unseen task type; for instance, we could evaluate the model on natural language inference (NLI) when no NLI tasks were seen during instruction tuning. Bottom: performance of zero-shot FLAN, compared with zero-shot and few-shot GPT-3, on three unseen task types where instruction tuning improved performance substantially out of ten we evaluate. NLI datasets: ANLI R1–R3, CB, RTE. Reading comprehension datasets: BoolQ, MultiRC, OBQA. Closed-book QA datasets: ARC-easy, ARC-challenge, NQ, TriviaQA.

LIMA: Less Is More for Alignment (1)

- ❑ An LLM's knowledge and reasoning capabilities are almost entirely learned during massive-scale unsupervised pre-training.
- ❑ **Instruction Tuning** and RLHF (more on this later) are not teaching the model *new* things.
- ❑ Instead, they are teaching the model the *superficial* format and style—the *persona* of a helpful assistant—to adopt when interacting.

LIMA: Less Is More for Alignment (2)

- ❑ High performance is achievable with minimal instruction data.
 - LIMA performed remarkably well compared to models trained on millions of examples (like DaVinci003).
 - Used only 1,000 meticulously curated prompt/response pairs.
- ❑ Quality > Quantity
 - A small, clean dataset of curated examples is vastly more efficient for alignment than a massive, messy one.
- ❑ Shifted the LLM Community's Focus
 - This paper popularized the idea of *data-efficient alignment* and spurred innovation in creating smaller, higher-quality datasets.